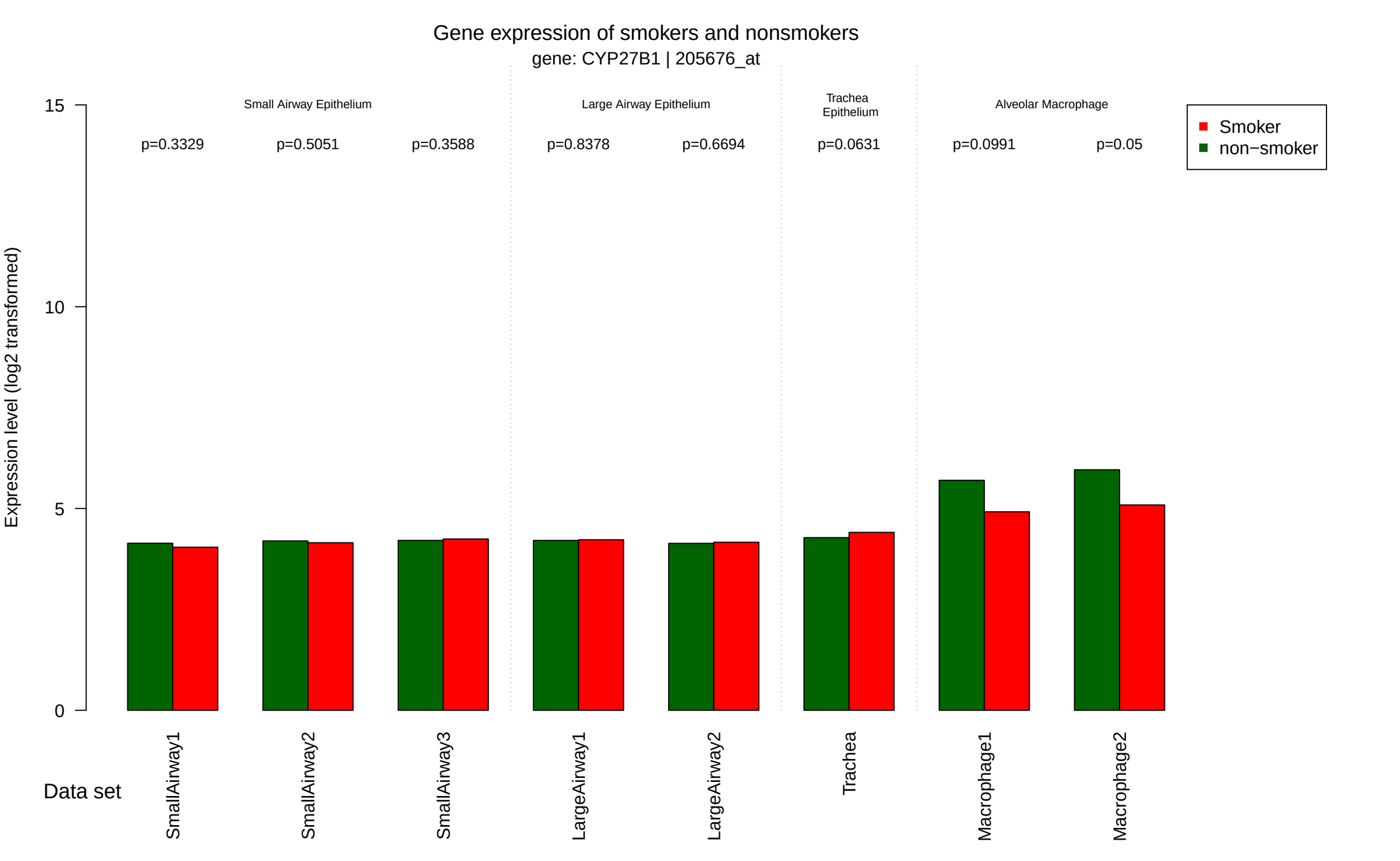
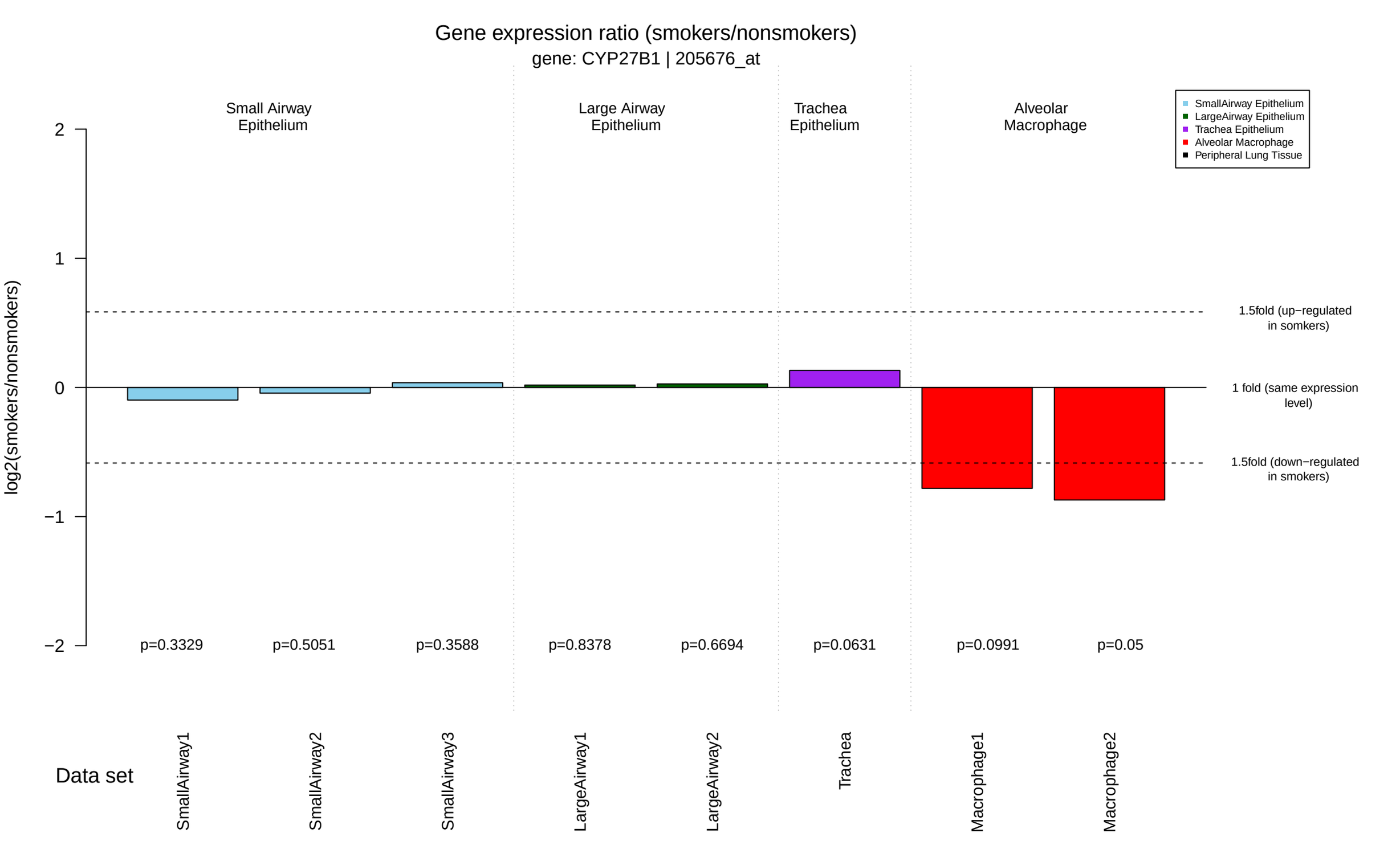
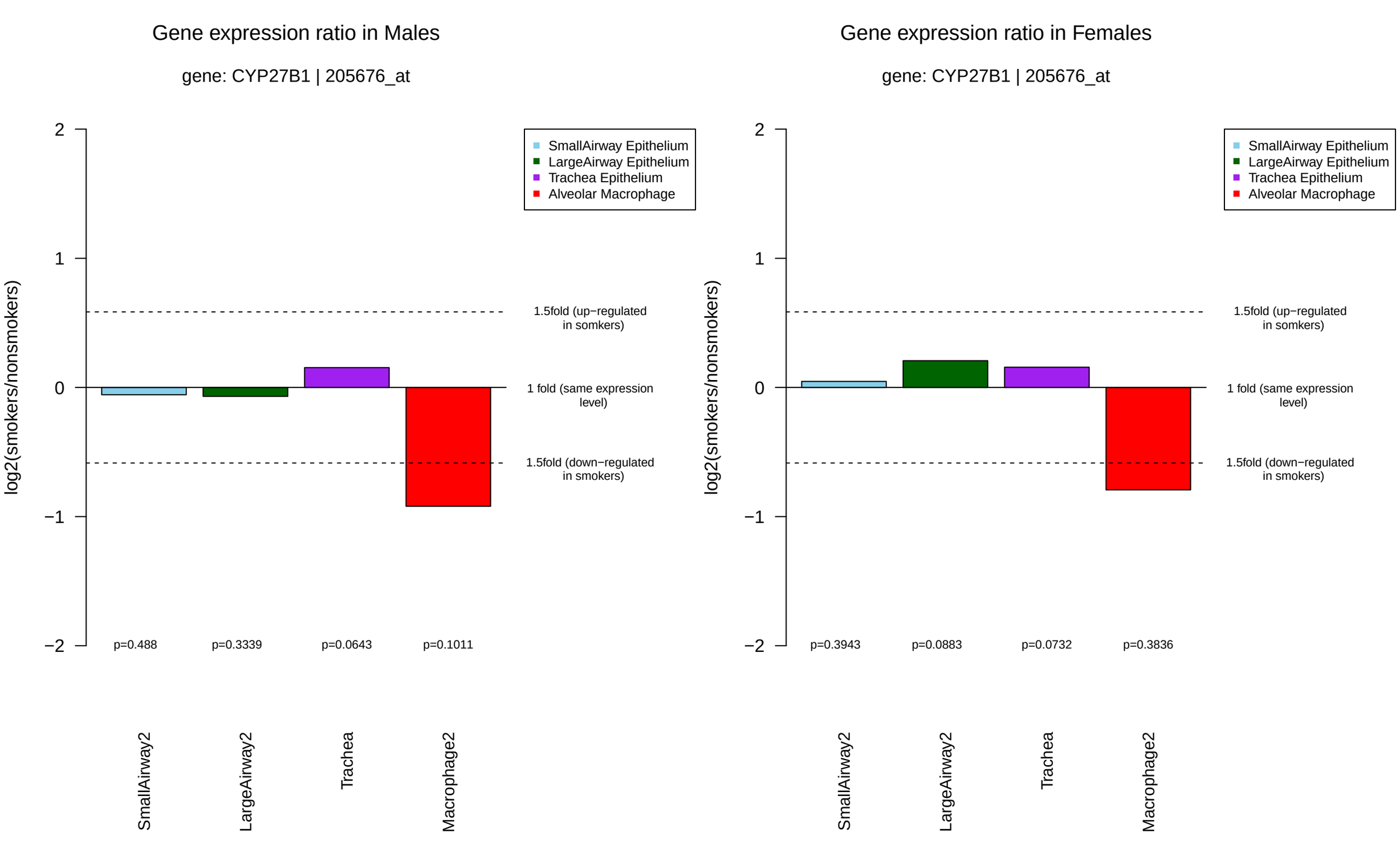
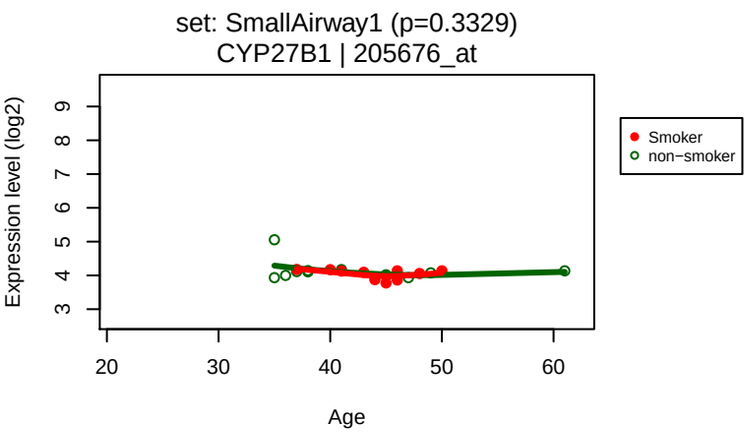
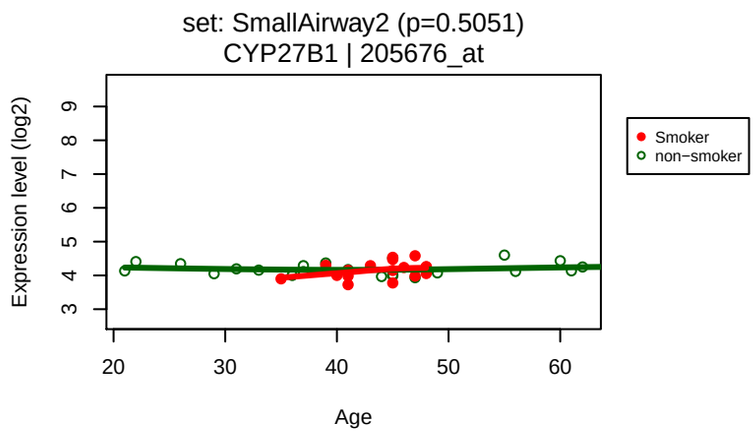
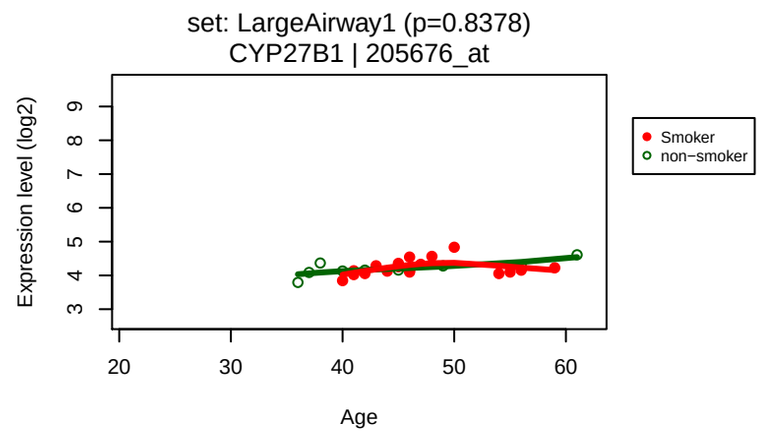

Gene expression of smokers and nonsmokers
gene: CYP27B1 | 205676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3329
p=0.5051
p=0.3588
p=0.8378
p=0.6694
p=0.0631
p=0.0991
p=0.05
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

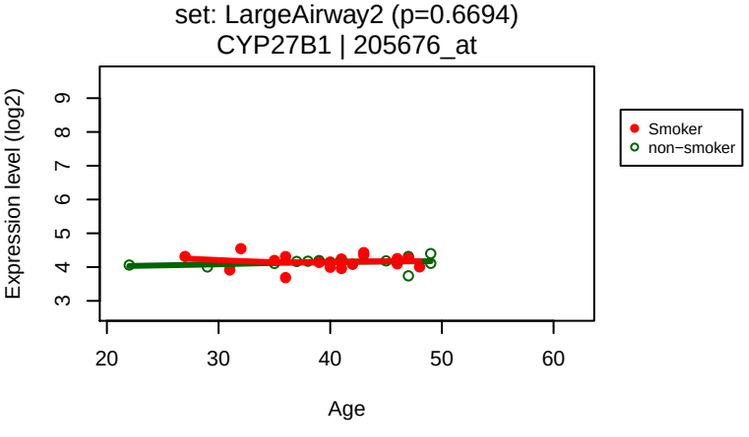

Gene expression ratio (smokers/nonsmokers)
gene: CYP27B1 | 205676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3329
p=0.5051
p=0.3588
p=0.8378
p=0.6694
p=0.0631
p=0.0991
p=0.05
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

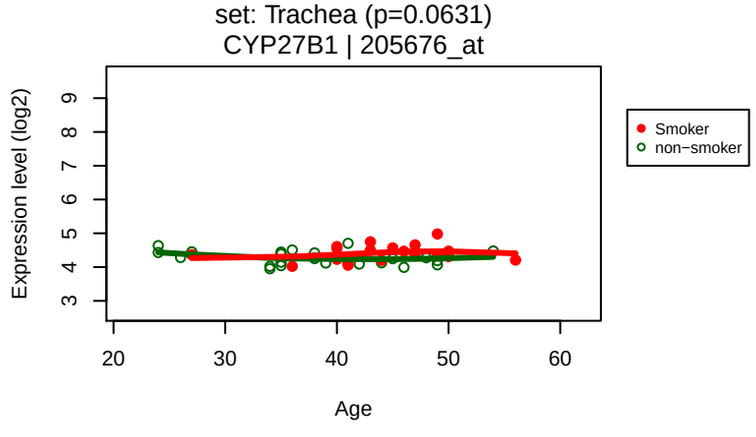

Gene expression ratio in Males
gene: CYP27B1 | 205676_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.488
p=0.3339
p=0.0643
p=0.1011
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP27B1 | 205676_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3943
p=0.0883
p=0.0732
p=0.3836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

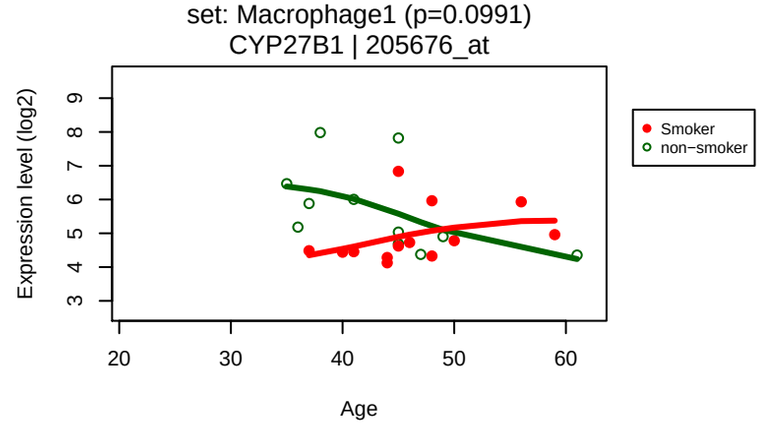

set: SmallAirway1 (p=0.3329)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker

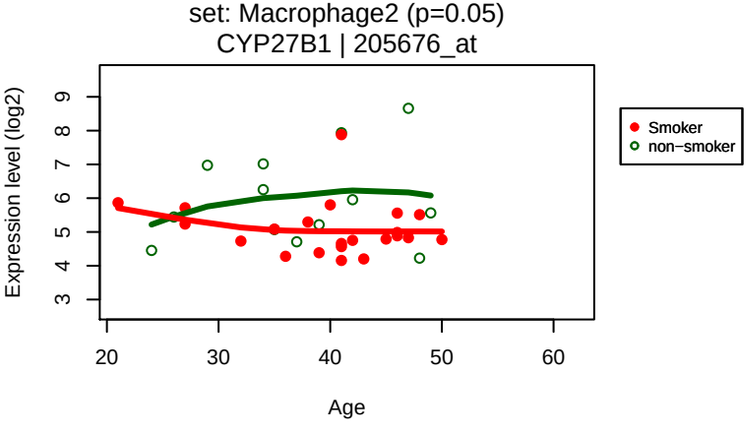

set: SmallAirway2 (p=0.5051)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker

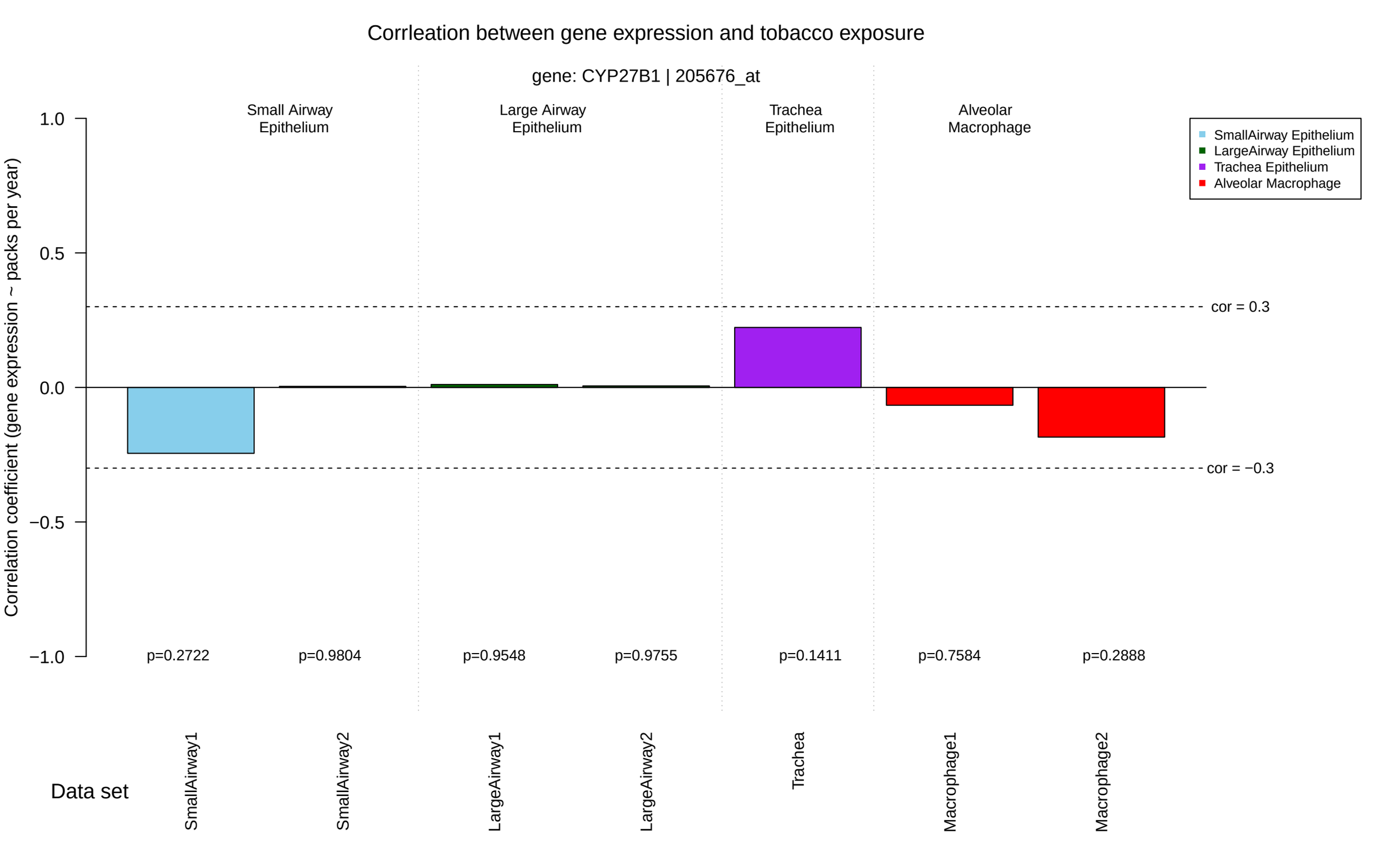

set: LargeAirway1 (p=0.8378)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6694)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0631)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0991)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.05)
CYP27B1 | 205676_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP27B1 | 205676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2722
p=0.9804
p=0.9548
p=0.9755
p=0.1411
p=0.7584
p=0.2888
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2